

Code-Layout, Readability and Reusability

Date	5 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Standard PEP 8 formatting is used across Python files; standard 2-space indentation is followed in HTML/CSS templates.
2	Proper File Structure	Files and folders are logically organized	Yes	Folders are logically segmented into backend code, dataset sources, trained model binaries, static resources and view layers.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names used
4	Function / Method Names	Functions are descriptively named	Yes	Clear and meaningful function names
5	Code Comments	Inline and block comments explain logic	Yes	Inline comments added wherever required

6	Modular Design	Code is split into reusable functions/modules	Yes	Separate notebooks and Flask modules
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable functions implemented
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic validation and exception handling implemented

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Model Training	High
2	Feature Scaling Module	Scikit-learn	Model Training & Prediction	High
3	Machine Learning Model	XGBoost / Random Forest	Flask Application	High
4	Prediction Function	Python	Flask Backend	High
5	HTML Templates	HTML, CSS	User Interface	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project follows a clean folder hierarchy with organized source files.
Readability	5	Meaningful naming conventions and comments improve understanding.
Reusability	5	Modular functions and reusable components reduce duplication.
Documentation / Comments	4.5	Important modules are documented with inline comments.
Overall Score	4.5	The project is well-structured, maintainable and suitable for future enhancements.